

Harsath Kaliappa Gounder Thangavel

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Education

York University

Bachelor of Science in Computer Science (Honours)

January 2020 – December 2024

Toronto, Canada

- Data structures
- Data mining
- Computer networking
- Computer architecture
- Algorithm design
- Database systems
- Compiler design
- Operating systems

Work Experience

Privex

Software Engineer

October 2023 – January 2024

London, United Kingdom

- Developed and deployed an AI customer service agent for Privex, a hosting company, using a self-hosted Large Language Model (LLM) and Retrieval Augmented Generation (RAG) with LangChain.
- Enabled instant responses to customer queries related to Privex products and services, reducing staff workload by handling approximately 70% of repeated common customer inquiries, significantly saving time and effort.
- Implemented REST APIs in Flask and integrated with Discord API to serve customers on Privex's Discord server.
- Utilized vector databases and MySQL for training data management, allowing dynamic updates from Discord by Privex staff.

CogniPet

Machine Learning Engineer

April 2019 – January 2020

Zürich, Switzerland

- Trained and deployed a deep learning model for recognizing 120 breeds of dogs using the Inception V3 neural network with 93% accuracy.
- Worked closely with a team of mobile and backend developers to implement a REST API and integrate it with CogniPet's Android and iOS platforms using Amazon Web Services (AWS).

Projects

Jix Interpreter | *Interpreter written in C*

- Developed an interpreter for a dynamically typed programming language with support for basic functional programming.
- Implemented features such as fault-tolerant parsing and error handling, dynamic types (int, bool, string, and arrays), functions, nested blocks, recursion, built-in functions, if-else, return, and break statements, as well as while and for loops.
- Added functional programming support, including first-class functions, higher-order functions, and function pointers.

HashRing | *Go module*

- Consistent hashing data structure implementation in Go using UUIDs and BLOBs, designed for load balancing and distributed systems.
- Implemented automated tests with GitHub Actions CI/CD pipelines.

Cap-Containers | *C header-only containers library*

- Highly-generic, self-contained, header-only containers written in C.
- Supports sets, maps, vectors, hash-tables (open and closed addressing), queues (circular, dynamic, fixed, and priority), stack, lists, and arena memory allocator.
- Includes thread-safe variants with cross-platform support (Windows, Linux, and macOS).

Blueth | *C++17 Linux networking framework*

- Networking framework written in C++17 for Linux and provides a set of common client-server abstractions for developing high-performance, concurrent, and event-driven networking services in a modular and composable manner.
- Features include event-loop abstraction, SOCKS5 client, HTTP parsers, proxies, thread-pool executors, caches, common codecs, SIMD wrappers, and networking utilities.

Volunteer Experience

Wikipedia

Editor

July 2021 – Present

- Contributed over 20,000 edits to Wikipedia, primarily focusing on topics within computer science.
- Elevated 4 data structure and cyber security articles to Good Article status.
- Collaborated with other editors to maintain the integrity and quality of the encyclopedia.

Technical Skills

Languages: Python, C++, C, Java, Go, JavaScript, PHP, SQL, $\text{\LaTeX}$, HTML/CSS, Bash, Perl
Technologies: Linux/Unix, MySQL/PostgreSQL, Django/Flask, JWTs, CI/CD pipelines, SMTP
Developer tools: Git, Docker, LXC/LXD, GDB/LLDB, Amazon Web Services